Dynamic programming-2
0-1 Knapsack problem

How to pack the bag to maximize total amount without overloading your luggage ?

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Reminder :Properties of a problem that can be solved with dynamic programming

Simple Subproblems

- We should be able to break the original problem to smaller subproblems that have the same structure

- Optimal Substructure
  - An optimal solution to a problem can be constructed efficiently from optimal solutions of its subproblems.
- Overlapping Subproblems
  - A problem has overlapping subproblems if finding its solution involves solving the same subproblem multiple times.

→ DP :Solving problems recursively by combining the solutions to similar smaller overlapping subproblems

Knapsack Problem

- We are given a set of n items, where each item i is specified by a weight w_i (or size s_i) and a value v_i.
- We are also given a limit W, the capacity of our knapsack.

<u>Problem</u>: How to pack the knapsack to achieve maximum total value of packed items?

Example:

n = 4 (# of elements)

W = 5 (maximum capacity)

Elements (weight, value) =

{ (1, 200), (3, 240), (2, 140), (5, 150) }

Knapsack Problem Types

There are two versions of the problem:

1. "0-1 knapsack problem"
   Items are indivisible; you either take an item or not. Some special instances can be solved with dynamic programming

2. "Fractional knapsack problem"
   Items are divisible : you can take any fraction of an item.

→0-1 means take item or leave it (no fractions).We are going to consider this problem.

The 0-1 Knapsack Problem

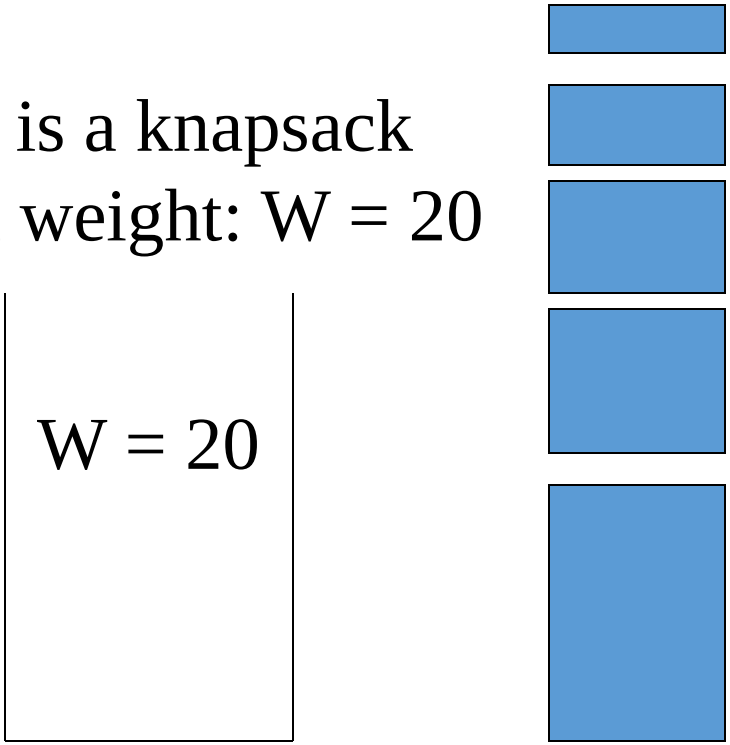
- How to apply Dynamic Programming ? We have difficulties.
- Difficulty is in characterizing subsets :
  - Find best solution for first k units and use it later – no good, as optimal solution doesn't necessarily build on earlier solutions. Why ?
    - Find best solution, for first k units, within quantity limit
    - Either use previous best at this limit, or new item plus previous best at reduced limit

  →Previous solution for k has to be changed.

  →This is contrary to standard Dynamic Programming !
  We need a different formulation of subsets.

0-1 Knapsack Problem: Example

This is a knapsack
Max weight: W = 20



Items	Weights w_i	Benefit values b_i
	2	3
	3	4
	4	5
	5	8
	9	10

0-1 Knapsack problem

Problem: Find

$$\max \sum_{i \in T} b_i \text{ subject to } \sum_{i \in T} w_i \leq W$$

- The problem is a *"0-1"* knapsack problem, because each item must be entirely accepted or rejected

Reminder :Solving 0-1 Knapsack Problem by Brute-force Approach

Let's first try to solve this problem with a straightforward Brute Force algorithm:

- Since there are *n* items, there are 2^n possible combinations of items.
- We go through all combinations and find the one with highest total value and with total size less than or equal to *W*
- Complexity will be $O(2^n)$

Example: n= 3, items A,B,C

Possible knapsack contents :

S={Empty, A,B, C, AB, AC, BC, ABC }

$|S| = 8 = 2^3$

Defining a Subproblem

- We can do better with an algorithm based on dynamic progamming
- We need to carefully identify the subproblems

DP solution should be composed of solutions to subproblems for subsets of k items (1,2,…,k).

→If items are labeled *1..n*, then a subproblem would be to find an optimal solution for

$S_k = \{items\ labeled\ 1, 2, ..\ k\}$

Defining a Subproblem

- This is a valid subproblem definition.
- But here dynamic programming question is: can we describe the final solution (S_n) in terms of subproblems (S_k) ?
- Unfortunately the answer is no, we can not do that. Why ?

Explanation follows….

Defining a Subproblem

Example:

- Assume that the best set of items for k=3: $S_k = \{I_0, I_1, I_2\}$ is $\{I_0, I_1, I_2\}$.
- The best set of items from $S_{k+1} = \{I_0, I_1, I_2, I_3\}$ may be $\{I_0, I_2, I_3\} \neq \{I_0, I_1, I_2\}$.
- In this example, the optimal solution, $\{I_0, I_2, I_3\}$, does NOT build upon the previous optimal solution.
- Instead it build's upon the solution $\{I_0, I_2\}$, which is really the optimal subset of $\{I_0, I_1, I_2\}$

Defining a Subproblem : Example

$w_1 = 2$ $b_1 = 3$	$w_2 = 4$ $b_2 = 5$	$w_3 = 5$ $b_3 = 8$	$w_4 = 3$ $b_4 = 4$	

Max weight: W = 20

Items in S_4: 1,2,3,4

Total weight: 14

Maximum benefit: 20

Item #	Weight w_i	Benefit b_i
1	2	3
2	4	5
3	5	8
4	3	4
5	9	10

S_4

S_5

$w_1 = 2$ $b_1 = 3$	$w_2 = 4$ $b_2 = 5$	$w_3 = 5$ $b_3 = 8$	$w_5 = 9$ $b_5 = 10$

Items in S_5 : 1,2,3,5

Total weight: 20

Maximum benefit: 26

Solution for S_4 is not part of the solution for S_5!!!

Adding another Parameter to Subproblems

- As we have seen, the solution for S_4 is not part of the solution for S_5
- So our definition of a subproblem is not valid and we need another one!
- Let's add another parameter: w, which will represent the <u>exact</u> total weight for each subset of items.

The problem is:

How to find an optimal solution for the combination (k,w)?

Recursive Formulation for subproblems

- The subproblem will then be to compute *V[k,w], i.e.,* to find an optimal solution for

  S_k = *{items labeled 1, 2, .. k} in a knapsack of size w*

Assuming we know V[i, j], where i=0,1, 2, … k-1, j=0,1,2, …w, how to derive V[k,w]?

V[k,w] = Table value for choices k and w.

→Find these values using previous entries in the table.

→This is Memoization!

Recursive Formulation for Subproblems

$$V[k,w] = \begin{cases} V[k-1,w] & \text{if } w_k > w \\ \max\{V[k-1,w], V[k-1,w-w_k]+b_k\} & \text{else} \end{cases}$$

The best subset of S_k that has the total weight $\leq w$, either contains item k or not. Two cases :

- First case: $w_k > w$. Item k can't be part of the solution, since if it was, the total weight would be $> w$, which is unacceptable.
- Second case: $w_k \leq w$. Then the item k <u>can</u> be in the solution, and we choose *the case with greater (max) value*.

Recursive Formulation for Subproblems:Explanation

The formulation means that, the best subset of S_k that has total weight w is one of the two:

1) the best subset of S_{k-1} that has total weight w, **or**

2) the best subset of S_{k-1} that has total weight w-w_k plus the value of item k

→The best subset of S_k that has the total weight w, either contains item k or not.

- First case: w_k> w. Item k can't be part of the solution, since if it was, the total weight would be > w, which is unacceptable
- Second case: w_k <= w. Then item k <u>can</u> be in the solution, and we choose the case with greater value

0-1 Knapsack Algorithm

Input: Sets of items S and w weights. Weights and values are stored in separate arrays. Output: Total value of best set items in a valid knapsack.

Algorithm Knapsack01

```
for (w = 0 to W)  V[0,w] = 0;
for (i = 1 to n    )  V[i,0] = 0;
for (i = 1 to n) {
  for (w = 0 to W) {
      if (w_i <= w)     // item i can be part of the solution
            if ( (b_i + V[i-1,w-w_i]) > V[i-1,w] )    V[i,w] = b_i + V[i-1,w- w_i]
            else (  V[i,w] = V[i-1,w] )
      else ( V[i,w] = V[i-1,w] )    // w_i > w
  }
}
```

First for-loop: We go through all the possible sizes of our knapsack until w and if item i is equal to 0, which is "V[0,w]", maximum value is 0. Because when i = 0, this means that we are not taking any item.

Second for-loop: i = 1 *to n*. We go through all the items from 1 to n and if the knapsack's size is equal to 0, which is "V[i, 0]". Corresponding values is again 0. Because when w = 0, this means that we can't put anything in the knapsack

The inner if and else conditions check if the current value is bigger than the previous value so as to maximize the values the knapsack can hold.

The outer if and else conditions inside the third for loop ($w_i \leq w$) check if the knapsack can hold the current item or not.

```
// A Dynamic Programming based c++ solution for 0-1 Knapsack

#include<iostream>
using namespace std;
int max(int a, int b) { return (a > b)? a : b; } // Returns the max value that can be put in a knapsack of capacity W

int knapSack(int W, int wt[], int val[], int n)
{
   int i, w;
   int K[n+1][W+1];    // Build table K[][] in bottom up manner

   for (i = 0; i <= n; i++) {
       for (w = 0; w <= W; w++) {
             if (i==0 || w==0)  K[i][w] = 0;
             else {
                     if (wt[i-1] <= w) K[i][w] = max(val[i-1] + K[i-1][w-wt[i-1]],  K[i-1][w]);
                     else K[i][w] = K[i-1][w];
              }
       }
   }
    return K[n][W];
}
```

```
int main()
{
    int val[] = {2,3,4,5};
    int wt[] = {3,4,5,6};
    int  W = 5;
    int n = sizeof(val)/sizeof(val[0]);
    //cout<<n<<endl;
    cout<<"Max. Knapsac Value = "<<knapSack(W, wt, val, n);
    return 0;
}

//Max. Knapsac Value = 7
```

Knapsack 0-1 Problem – Run Time

```
for w = 0 to W
  B[0,w] = 0
```

$O(W)$

```
for i = 1 to n
  B[i,0] = 0
```

$O(n)$

```
for i = 1 to n
  for w = 0 to W
      < the rest of the code >
```

Repeat n times

$O(W)$

The running time of this algorithm is

$T(n) = O(W) + O(n) + O(n*W)$

$= O(n*W)$

Run Time Comparison

- Remember that the brute-force algorithm takes $O(2^n)$
- When comparing complexities, we find that depending on W, either the dynamic programming algorithm is more efficient or the brute force algorithm could be more efficient.
- For example, for n=5, W=100000, brute force is preferable, but for n=30 and W=1000, the dynamic programming solution is preferable:

Case1: n=5 : $2^n = 32$, n*W = 500000

Case2: n=30 : $2^n = 1073741824$, n*W= 30000

Tracing 0-1 Knapsack Algorithm : Example(1)

Consider the following sample data:

Input :

n = 4 (# of elements)

W = 5 (max weight,capacity)

Elements : (weight, benefit)

(2,3), (3,4), (4,5), (5,6)

Output :

The table (matrix) V

Example (2)

Initialize first row and column to 0.

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1						
2						
3						
4						

for w = 0 to W

V[0,w] = 0

Example (3)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0					
2	0					
3	0					
4	0					

```
for i = 1 to n
        V[i,0] = 0
```

Example (4)

Items:

1: (2,3)

2: (3,4)

3: (4,5)

4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0				
2	0					
3	0					
4	0					

$i=1$

$b_i=3$

$w_i=2$

$w=1$

$w-w_i=-1$

if w_i <= w // item i can be part of the solution

if b_i + V[i-1,w-w_i] > V[i-1,w]

V[i,w] = b_i + V[i-1,w- w_i]

else

V[i,w] = V[i-1,w]

else V[i,w] = V[i-1,w] // w_i > w , so accept this.

Example (5)

Items:

1: (2,3)

2: (3,4)

3: (4,5)

4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	**3**			
2	0					
3	0					
4	0					

$i=1$

$b_i=3$

$w_i=2$

$w=2$

$w-w_i=0$

if w_i <= w // item i can be part of the solution

if b_i + V[i-1,w-w_i] > V[i-1,w] // Yes

V[i,w] = b_i + V[i-1,w- w_i]

else

V[i,w] = V[i-1,w]

else V[i,w] = V[i-1,w] // w_i > w

Example (6)

Items:

1: (2,3)

2: (3,4)

3: (4,5)

4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	**3**		
2	0					
3	0					
4	0					

$i=1$

$b_i=3$

$w_i=2$

$w=3$

$w-w_i=1$

if $w_i <= w$ // item i can be part of the solution

if $b_i + V[i-1,w-w_i] > V[i-1,w]$ // yes

$V[i,w] = b_i + V[i-1,w- w_i]$

else

$V[i,w] = V[i-1,w]$

else $V[i,w] = V[i-1,w]$ // $w_i > w$

Example (7)

Items:

1: (2,3)

2: (3,4)

3: (4,5)

4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	**3**	
2	0					
3	0					
4	0					

$i=1$

$b_i=3$

$w_i=2$

$w=4$

$w-w_i=2$

if w_i <= w // item i can be part of the solution

if $b_i + V[i-1,w-w_i] > V[i-1,w]$ // Yes

$V[i,w] = b_i + V[i-1,w- w_i]$

else

$V[i,w] = V[i-1,w]$

else $V[i,w] = V[i-1,w]$ // $w_i > w$

Example (8)

Items:

1: (2,3)

2: (3,4)

3: (4,5)

4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	**3**
2	0					
3	0					
4	0					

$i=1$

$b_i=3$

$w_i=2$

$w=5$

$w-w_i=3$

if w_i <= w // item i can be part of the solution

if $b_i + V[i-1,w-w_i] > V[i-1,w]$ // Yes

$V[i,w] = b_i + V[i-1,w- w_i]$

else

$V[i,w] = V[i-1,w]$

else $V[i,w] = V[i-1,w]$ // $w_i > w$

Example (9)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	**0**				
3	0					
4	0					

$i=2$

$b_i=4$

$w_i=3$

$w=1$

$w-w_i=-2$

if w_i <= w // item i can be part of the solution
    if b_i + V[i-1,w-w_i] > V[i-1,w]
        V[i,w] = b_i + V[i-1,w- w_i]
    else
        V[i,w] = V[i-1,w]
else V[i,w] = V[i-1,w] // w_i > w

Example (10)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	**3**			
3	0					
4	0					

$i=2$

$b_i=4$

$w_i=3$

$w=2$

$w-w_i=-1$

if w_i <= w // item i can be part of the solution
    if b_i + V[i-1,w-w_i] > V[i-1,w]
        V[i,w] = b_i + V[i-1,w- w_i]
    else
        V[i,w] = V[i-1,w]
else V[i,w] = V[i-1,w] // w_i > w

Example (11)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	**4**		
3	0					
4	0					

$i=2$

$b_i=4$

$w_i=3$

$w=3$

$w-w_i=0$

if w_i <= w // item i can be part of the solution

if $b_i + V[i-1,w-w_i] > V[i-1,w]$

$V[i,w] = b_i + V[i-1,w- w_i]$

else

$V[i,w] = V[i-1,w]$

else $V[i,w] = V[i-1,w]$ // $w_i > w$

Example (12)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	**4**	
3	0					
4	0					

$i=2$

$b_i=4$

$w_i=3$

$w=4$

$w-w_i=1$

if w_i <= w // item i can be part of the solution

if b_i + V[i-1,w-w_i] > V[i-1,w]

V[i,w] = b_i + V[i-1,w- w_i]

else

V[i,w] = V[i-1,w]

else V[i,w] = V[i-1,w] // w_i > w

Example (13)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0					
4	0					

$i=2$

$b_i=4$

$w_i=3$

$w=5$

$w-w_i=2$

if $w_i <= w$ // item i can be part of the solution

if $b_i + V[i-1,w-w_i] > V[i-1,w]$

$V[i,w] = b_i + V[i-1,w- w_i]$

else

$V[i,w] = V[i-1,w]$

else $V[i,w] = V[i-1,w]$ // $w_i > w$

Example (14)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	**0**	**3**	**4**		
4	0					

i=3

b_i=5

w_i=4

w= 1..3

```
if w_i <= w // item i can be part of the solution
    if b_i + V[i-1,w-w_i] > V[i-1,w]
        V[i,w] = b_i + V[i-1,w- w_i]
    else
        V[i,w] = V[i-1,w]
else V[i,w] = V[i-1,w]  // w_i > w
```

Example (15)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	**5**	
4	0					

$i=3$

$b_i=5$

$w_i=4$

$w= 4$

$w- w_i=0$

if w_i <= w // item i can be part of the solution

if b_i + V[i-1,w-w_i] > V[i-1,w]

V[i,w] = b_i + V[i-1,w- w_i]

else

V[i,w] = V[i-1,w]

else V[i,w] = V[i-1,w] // w_i > w

Example (16)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0					

$i=3$

$b_i=5$

$w_i=4$

$w= 5$

$w- w_i=1$

```
if w_i <= w // item i can be part of the solution
    if b_i + V[i-1,w-w_i] > V[i-1,w]
        V[i,w] = b_i + V[i-1,w- w_i]
    else
        V[i,w] = V[i-1,w]
else V[i,w] = V[i-1,w]  // w_i > w
```

Example (17)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	**0**	**3**	**4**	**5**	

i=4

b_i=6

w_i=5

w= 1..4

```
if w_i <= w // item i can be part of the solution
    if b_i + V[i-1,w-w_i] > V[i-1,w]
        V[i,w] = b_i + V[i-1,w- w_i]
    else
        V[i,w] = V[i-1,w]
else V[i,w] = V[i-1,w]  // w_i > w
```

Example (18)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

$i=4$

$b_i=6$

$w_i=5$

$w= 5$

$w- w_i=0$

```
if w_i <= w // item i can be part of the solution
    if b_i + V[i-1,w-w_i] > V[i-1,w]
        V[i,w] = b_i + V[i-1,w- w_i]
    else
        V[i,w] = V[i-1,w]
else V[i,w] = V[i-1,w]  // w_i > w
```

Comments

- The maximum value of benefit that can be obtained for optimal packing is 7.
- This algorithm only finds the max possible value (Total benefit)that can be carried in the knapsack
- To know the items that make this maximum value, an addition to this algorithm is necessary
- All we need is to trace back the computations for the maximum value in the table

How to find actual Knapsack Items

- All of the information we need is in the table.
- $V[n,W]$ is the maximal value of items that can be placed in the Knapsack.
- Let i=n and k=W

if $V[i,k] \neq V[i-1,k]$ then

  mark the ith item as in the knapsack

  $i = i-1$, $k = k-w_i$

else

  $i = i-1$ // Assume the ith item is <u>not</u> in the knapsack

  // Could it be in the optimally packed knapsack?

Finding the Items

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=4

k= 5

b_i=6

w_i=5

$V[i,k] = 7$

$V[i-1,k] = 7$

i=n, k=W

while i,k > 0

if $V[i,k] \neq V[i-1,k]$ then

mark the ith item as in the knapsack

$i = i-1$, $k = k-w_i$

else

$i = i-1$

Finding the Items (2)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=4

k= 5

b_i=6

w_i=5

$V[i,k] = 7$

$V[i-1,k] = 7$

i=n, k=W

while i,k > 0

if $V[i,k] \neq V[i-1,k]$ then

mark the ith item as in the knapsack

$i = i-1$, $k = k-w_i$

else

$\boldsymbol{i = i-1}$

Finding the Items (3)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=3

k= 5

b_i=5

w_i=4

$V[i,k] = 7$

$V[i-1,k] = 7$

i=n, k=W

while i,k > 0

if $V[i,k] \neq V[i-1,k]$ then

mark the ith item as in the knapsack

$i = i-1$, $k = k-w_i$

else

$\boldsymbol{i = i-1}$

Finding the Items (3)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=3

k= 5

b_i=5

w_i=4

$V[i,k] = 7$

$V[i-1,k] = 7$

i=n, k=W

while i,k > 0

if $V[i,k] \neq V[i-1,k]$ then

mark the ith item as in the knapsack

$i = i-1$, $k = k-w_i$

else

$\boldsymbol{i = i-1}$

Finding the Items (4)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=2

k= 5

b_i=4

w_i=3

$V[i,k] = 7$

$V[i-1,k] = 3$

$k - w_i = 2$

```
i=n, k=W
while i,k > 0
    if V[i,k] ≠ V[i–1,k] then
        mark the ith item as in the knapsack
        i = i–1, k = k-w_i
    else
        i = i–1
```

Finding the Items (5)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=1

k= 2

b_i=3

w_i=2

$V[i,k] = 3$

$V[i-1,k] = 0$

$k - w_i = 0$

i=n, k=W

while i,k > 0

if $V[i,k] \neq V[i-1,k]$ then

mark the ith item as in the knapsack

$i = i-1,\ k = k-w_i$

else

$\boldsymbol{i = i-1}$

Finding the Items (6)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=0
k= 0

The optimal knapsack should contain {1, 2}

i=n, k=W
while i,k > 0
    if $V[i,k] \neq V[i-1,k]$ then
        mark the nth item as in the knapsack
        $i = i-1$, $k = k-w_i$
    else
        $i = i-1$

Finding the Items (7)

Items:

1: (2,3)
2: (3,4)
3: (4,5)
4: (5,6)

i\W	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

i=n, k=W // No,stop
while i,k > 0
    if $V[i,k] \neq V[i-1,k]$ then
        mark the nth item as in the knapsack
        $i = i-1$, $k = k-w_i$
    else
        $i = i-1$

The optimal knapsack should contain {1, 2}

Knapsack Problem : Applications

- Applicable in most resource allocation problems where there are financial or other constraints :
  - Resource allocation with financial constraints
  - Construction and scoring of heterogeneous test sets (If each question has a different value , how to maximize the mark?)
  - Selection of capital investments
- Finding the least wasteful way to cut raw materials.
- Computer games : Pick the most valuable set of items from a treasure to carry.

Dynamic Programming : Conclusion

- Dynamic programming is a technique for making a sequence of interrelated decisions.
- "Programming": A tabular method (not writing computer code)
- It may be used to solve Dynamic optimization problems, where the problem can be split into overlapping subproblems
- It requires formulating an appropriate recursive relationship for each individual subproblem
- When the solution can be recursively described in terms of optimal partial solutions, we can store these partial solutions and re-use them as necessary(Memoization).
- Running times are usullly better than Brute Force and other naïve algorithms . Example:

0-1 Knapsack problem: $O(W*n)$ vs. $O(2^n)$